

Uppsala University
Department of Information Technology

Literature Seminar Report

Course: Data Engineering II

Arnab Kumar Ghosh
Eric Buxton
Prodip Kumar Das
Rahul Bhat

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1 Q1: What is the meaning and importance of contextualization and orchestration, and what are currently available tools and services in this area?

Role and Importance

Contextualization is the automated process of setting up a new virtual machine or container so it can start working correctly [1]. When cloud systems scale up, they create new copies from a standard template. Contextualization injects specific details like IP addresses and security keys into these copies so they have their own identity [1].

Orchestration is the broader management of these parts. It coordinates how different services work together across many computers and locations [6]. It is important because it automates complex tasks like deploying software and managing resources, which reduces human error and keeps the system running smoothly [1, 6].

Four Frameworks and Tools

1. **Kubernetes (Orchestration):** This tool manages containerized applications. It automatically handles scaling and ensures that if one part fails, another starts immediately to keep the service online [4].
2. **Puppet and Chef (Configuration Management):** These tools automate the setup and maintenance of software on many virtual machines at once. They make sure every machine follows the same rules and stays updated [1].
3. **VM Contextualizer (Contextualization):** This is a specific tool that prepares virtual machine images by injecting custom data during boot time. It helps keep the main image simple while allowing each copy to be unique [1].
4. **TOSCA (Orchestration):** This is a standard way to describe how a cloud application should be built. It tells orchestration tools exactly what hardware and connections are needed to run a complex application [6].

2 Q2: How does contemporary serverless architecture relate to the digitalization of society and Big Data applications?

Contemporary serverless architecture has become an important part of the digitalization of society because it allows organizations to create and run scalable digital services without having to manage physical servers or complicated infrastructure themselves [2]. In serverless computing, cloud providers automatically handle resources and run functions only when they are needed. This approach supports modern services such as online banking, e-commerce, IoT systems, streaming platforms, and AI applications, where user activity can change quickly. Since companies only pay for the resources they actually use, serverless computing can reduce costs and make innovation easier and more accessible [2, 5].

Serverless architecture is also strongly connected to Big Data applications. Today's Big Data systems generate huge amounts of real-time information from mobile devices, sensors, social media, and online transactions [3]. Serverless platforms can automatically scale up or down to process this data efficiently. Technologies such as AWS Lambda, Google Cloud Functions, and Azure Functions are often combined with Big Data tools like Apache Kafka, Spark, and cloud storage systems to support real-time analytics and event-driven data processing [2, 5]. Because of this, serverless solutions are especially useful for recommendation systems, fraud detection, and smart city technologies.

Another major benefit is faster software development and deployment. Developers can spend

more time focusing on the application itself instead of maintaining servers, which improves flexibility and supports continuous updates [5]. At the same time, serverless systems also have some disadvantages, such as startup delays, dependence on specific cloud providers, and difficulties when debugging distributed systems [2]. Even with these challenges, serverless computing is increasingly seen as an important technology for supporting both the digital transformation of society and the growing need for scalable Big Data processing.

3 Q4: Explain the role and importance of model serving and CI/CD in today's data engineering world. Write down the names and briefly explain four frameworks (two for model serving and two for CI/CD) that offer model serving and CI/CD capabilities.

Role and Importance

Training a model is only the first step. To be useful, the model must run as a live service so other applications can send data and receive predictions — this is model serving. It manages model versions, routes requests, and scales with demand. CI/CD (Continuous Integration and Continuous Deployment) automates the steps between updating a model and deploying it to production. Because data changes over time, models need regular retraining and redeployment. Without CI/CD, this is slow and error-prone. Together, model serving and CI/CD let teams update models quickly, catch errors through automated testing, and keep predictions stable in production [7, 8].

Four Frameworks

1. **TensorFlow Serving (Model Serving):** An open-source tool by Google for deploying TensorFlow models. It serves predictions via REST or gRPC APIs and supports multiple model versions at once, which is useful for A/B testing. It is built for high-speed, high-load environments.
2. **MLflow Model Serving (Model Serving):** An open-source platform that works with any ML framework. A single command deploys a trained model as a REST API. It also provides a model registry to track versions and safely move models from testing to production.
3. **Jenkins (CI/CD):** One of the most widely used CI/CD tools. It can automatically retrain a model when new data arrives, run validation tests, and deploy passing models. Its many plugins connect it to cloud platforms, Docker, and ML frameworks.
4. **PerfCI (CI/CD):** A CI/CD framework designed specifically for ML pipelines. It automatically benchmarks model performance such as speed and accuracy. Each time a change is made, it flags regressions before the model reaches production. This makes it especially useful for teams that need to ensure model quality does not drop between updates.

4 Q5: What is the difference between distributed and Federated machine learning? Highlight the architectural differences between the two machine learning environments. Which one of them is more suitable for privacy preserving model training process and why?

Distributed Machine Learning (DML) and Federated Learning (FL) are two fundamentally different approaches to training machine learning models across multiple computers [Verbraeken2020]. DML centralizes data in one location, then splits the training work across multiple computers. Data can be moved between computers if needed, and the main goal is to speed up training

on huge datasets by processing them in parallel [Verbraeken2020]. In contrast, Federated Learning keeps data spread across different devices and it never moves to a central server [Yang2021]. Instead, each device trains the model locally on its own data, then sends only the updated model weights back to the server. The server then combines all the updates and sends the improved global model back to the devices.

4.1 Architectural Differences

In DML, data flows freely between computers and the central server controls and manages everything directly [Verbraeken2020]. The system is designed to maximize processing speed and computational power. In FL, data stays permanently on local devices and never travels anywhere [Yang2021]. Instead of data moving, only small model updates are exchanged. The server's role changes from being a controlling authority to simply coordinating and combining updates from all participants [Qian2020]. This fundamental difference in architecture creates very different privacy profiles for the two systems.

4.2 Privacy Considerations

When considering privacy-preserving model training, Federated Learning is clearly the superior choice [Yang2021]. The most important reason is that raw personal data never leaves the user's device. In DML, data moves around between computers and can be exposed if central servers get hacked [Verbraeken2020]. In FL, hackers would need to break into many individual devices to steal data, rather than just compromising one central server. Additionally, privacy laws like GDPR require data to stay under user control and not be centralized [Yang2021]. FL naturally follows these rules because data never leaves home.

For example, a hospital using FL can train a disease detection model with other hospitals without ever sharing patient records [Qian2020]. With DML, all sensitive patient data would need to go to one central server, creating an enormous privacy risk. Both systems can add extra security measures like encryption, but FL is privacy-first by design, not just as an add-on feature [Yang2021].

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